

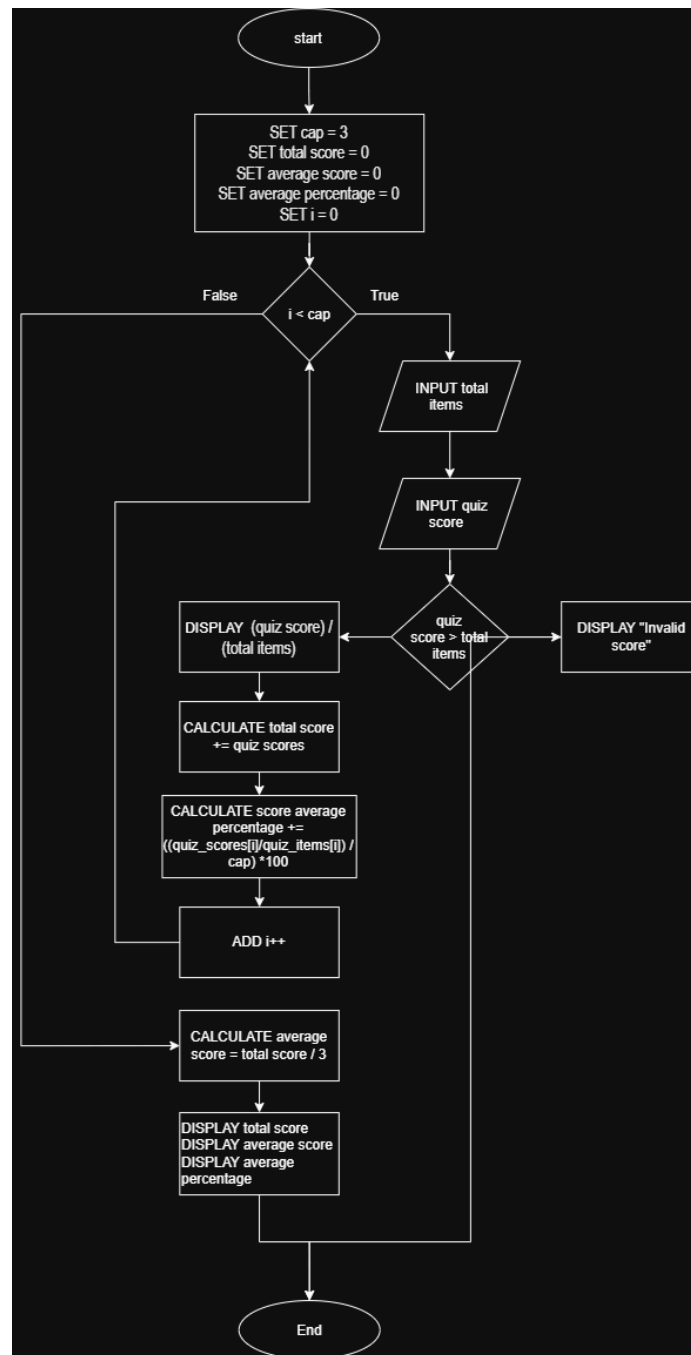
COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	SECTION:	CPE21S1
DATE PERFORMED:	Jul 8, 2026	DATE SUBMITTED:	Aug 13, 2026
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ACTIVITY NO. 1.1 Algorithm, Flowchart, Pseudocode

1. PROCEDURE OUTPUT

1. Compute the total and average of 3 quizzes then display the total and average.	
Algorithm	Start Set the maximum quizzes to 3 (cap). Set total_score and score_average to 0. Repeat for each of the 3 quizzes: Enter the total number of items for the quiz. Enter the score obtained. Check if the score is greater than the total items. If yes, display "Invalid score" and stop. Display the quiz score. Add the score to the total_score. Calculate the quiz percentage and add it to score_average. Display the total scores. Display the average score. Display the average score percentage. End

Flowchart



Pseudocode

```

START
SET cap = 3
SET total score = 0
SET average score = 0
SET average percentage = 0
REPEAT for 3 quizzes:

```

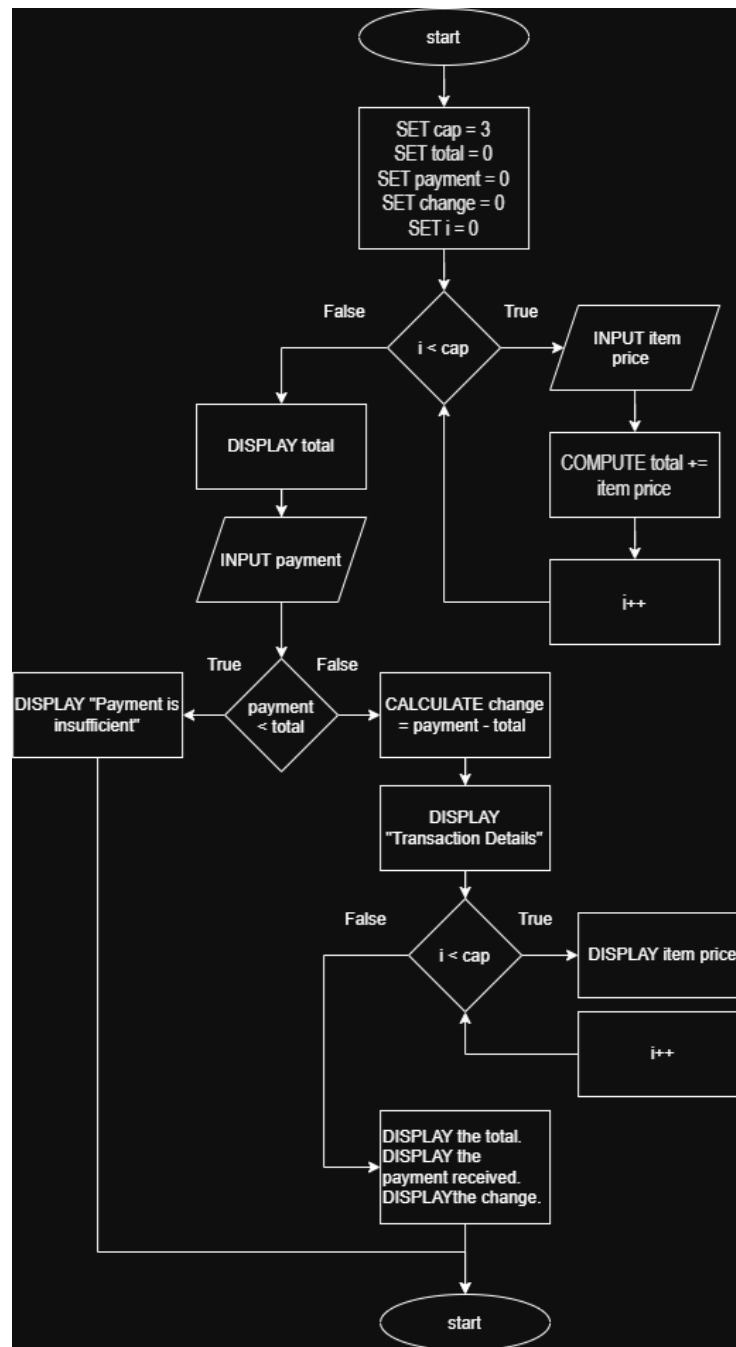
	<pre> INPUT total items INPUT quiz score IF quiz score > total items: DISPLAY "Invalid score" END PROGRAM DISPLAY (quiz score) / (total items) CALCULATE total score += quiz scores CALCULATE score average percentage += ((quiz_scores[i]/quiz_items[i]) / cap) *100 CALCULATE average score = total score / 3 ADD i++ DISPLAY total score DISPLAY average score DISPLAY average percentage END </pre>
Code	<pre> 1 #include <iostream> 2 3 int main() { 4 5 const int cap = 3; 6 float quiz_items[cap] = {}; 7 float quiz_scores[cap] = {}; 8 float total_score = 0, score_average = 0, score_average_p = 0; 9 int i; 10 11 for (i = 0; i < cap; i++) { 12 std::cout << "Enter total items of Quiz " << i+1 << ": "; 13 std::cin >> quiz_items[i]; 14 15 std::cout << "Quiz " << i+1 << " score: "; 16 std::cin >> quiz_scores[i]; 17 18 if(quiz_scores[i] > quiz_items[i]) { 19 std::cout << "invalid score"; 20 return 0; 21 } 22 23 std::cout << "Score: " << quiz_scores[i] << " / " << quiz_items[i] << std::endl; 24 total_score += quiz_scores[i]; 25 score_average_p += ((quiz_scores[i]/quiz_items[i]) / cap) *100; 26 27 } 28 29 std::cout << std::endl; 30 31 score_average = total_score / 3; 32 33 std::cout << "Total scores: " << total_score << std::endl; 34 std::cout << "Average score: " << score_average << std::endl; 35 std::cout << "Average score %: " << score_average_p << "%" << std::endl; 36 37 38 return 0; 39 } </pre>

Code Output	<pre> Enter total items of Quiz 1: 10 Quiz 1 score: 5 Score: 5 / 10 Enter total items of Quiz 2: 20 Quiz 2 score: 15 Score: 15 / 20 Enter total items of Quiz 3: 30 Quiz 3 score: 20 Score: 20 / 30 Total scores: 40 Average score: 13.3333 Average score %: 63.8889% </pre>
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2. Cashiering system that computes for total price of the 3 items bought, accept payment and computes change. Display the transactions made.

Algorithm	<pre> Start Set cap to 3. Set total, payment, and change to 0. Repeat the following steps for 3 items: Input the item price. Add the item price to total. Display the total price. Input the payment amount. If the payment is less than the total: Display "Payment is insufficient". End the program. Calculate change = payment - total. Display "Transaction Details". Repeat for the 3 items: Display the item price. Display the total. Display the payment received. Display the change. End. </pre>
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Flowchart



Pseudocode

START
 SET cap = 3
 SET total = 0
 SET payment = 0
 SET change = 0
 REPEAT for 3 items:

```
INPUT item price
COMPUTE total += item price
DISPLAY total
INPUT payment amount

IF payment < total THEN
    DISPLAY "Payment is insufficient"
    END PROGRAM
END IF

CALCULATE change = payment - total

DISPLAY "Transaction Details"

REPEAT for 3 items:
    DISPLAY item price

DISPLAY total
DISPLAY payment received
DISPLAY change

END
```

Code

```
1  #include <iostream>
2
3  int main() {
4
5      const int cap = 3;
6      float items[cap] = {};
7      float total = 0, change = 0, payment = 0;
8      int i = 0;
9
10     for(i = 0; i < cap; i++){
11         std::cout << "Item " << i+1 << " price: ";
12         std::cin >> items[i];
13
14         total += items[i];
15     }
16
17     std::cout << "Total price: " << total << std::endl;
18     std::cout << "Enter payment amount: ";
19     std::cin >> payment;
20
21     if (payment < total) {
22         std::cout << "Payment is insufficient";
23         return 0;
24     }
25
26     change = payment - total;
27
28     std::cout << "=====\n";
29     std::cout << "    Transaction Details" << std::endl;
30     std::cout << "=====\n\n";
31
32     for(i = 0; i < cap; i++){
33         std::cout << "Item " << i+1 << "price: " << items[i] << std::endl;
34     }
35
36     std::cout << std::endl << "Total: " << total << std::endl;
37     std::cout << "Payment received: " << payment << std::endl;
38     std::cout << "Change: " << change << std::endl;
39
40     return 0;
41
42 }
```

Code Output

```
Item 1 price: 10
Item 2 price: 20
Item 3 price: 30
Total price: 60
Enter payment amount: 70
=====
Transaction Details
=====

Item 1price: 10
Item 2price: 20
Item 3price: 30

Total: 60
Payment reveived: 70
Change: 10
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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